

Megh Patel

📞 9510983397 | ✉️ patelmeghmahesh2701@gmail.com | 📍 Ahmedabad, India
🌐 meghpatel2711 | 🌐 meghpatel.vercel.app | 🔄 MeGH2711

ABOUT ME

Focused and analytical, I am a keen and continuous learner dedicated to professional and personal growth. I approach challenges as an analytical thinker and decisive problem-solver, looking beyond a purely technical lens to incorporate user and client-centric observations into my decision-making process. My mindset is driven by the ability to balance technical logic with real-world impact, ensuring that every solution is both robust and purposefully aligned with the needs of the end-user.

WORK EXPERIENCE

Elkay Software Pvt. Ltd.

Feb 2025 - Apr 2025

Web Developer & Designer Intern

Ahmedabad, India

- Built responsive UI using HTML, CSS, JS, Kendo UI & Bootstrap, improving load time by 30%.
- Led integration of APIs for accurate data display.
- Collaborated in Agile sprints, ensuring 100% on-time delivery.
- Learnt WCAG 3.0 compliance to support health-tech tasks, accelerating onboarding by 50%.

360 Design

Jun 2024 – July 2024

Web Developer Intern

(Remote) Ahmedabad, India

- Worked on web design using React.js, focusing on building interactive UIs and managing state with hooks.
- Developed CLOTHIFY, an online clothing store using the MERN stack, featuring product categorization and CRUD operations via RESTful APIs.
- Gained experience integrating APIs with Axios and enhancing app scalability.

Gandhinagar University

Jan 2024 - Mar 2024

TechFest Organizer & Host

Gandhinagar, India

- Led end-to-end coordination for large-scale campus tech events, engaging over 3,000 attendees.
- Hosted multiple coding and gaming competitions and workshops.
- Contributed to event branding, marketing strategy, and enhancing the overall attendee experience.

Oasis Infobyte

Mar 2023 - Apr 2023

Web Development & Designing Intern

(Remote) Ahmedabad, India

- Gained hands-on experience in web development by working on real-world projects.
- Quickly adapted to using modern frameworks like Bootstrap, enhancing front-end development skills.
- Built a professional portfolio showcasing projects and skills effectively.

PROJECTS

HIE Detection using Deep Learning

🔗 [github/hiedetection](#)

- Developed an automated detection system for **Hypoxic-Ischemic Encephalopathy (HIE)** using deep learning to analyze neonatal MRI scans for early diagnosis.
- Implemented and **fine-tuned Swin-Unet** to identify subtle patterns of brain injury, achieving high sensitivity and specificity in segmentation.
- Pre-processed and augmented clinical neuroimaging dataset to improve model robustness and generalize performance across different MRI scenarios.

- Evaluated model performance using metrics such as Dice Coefficient and Intersection over Union, demonstrating significant improvement over traditional manual assessment methods.
- Achieved **Test Mean IoU as 0.421** and **Test Mean Dice as 0.542**, which is not actually sufficient for the medical segmentation domain, but there is still a wide scope of improvement.

Road Marking Segmentation for Autonomous Systems

 [github/roadmarkingsegmentation](#)

- Developed a semantic segmentation pipeline using PyTorch and MMSegmentation to identify 13 categories of road markings (e.g., Zebra crossings, lane lines, arrows) on the CeyMo dataset.
- Implemented and compared DeepLabV3+ (CNN-based with ASPP) and SegFormer (Transformer-based) architectures to evaluate local spatial precision versus global context awareness.
- Engineered an Ensemble Model using a weighted soft-voting mechanism on raw probabilistic maps (logits) to reduce individual model errors and false negatives.
- Optimized training through heavy data augmentation including Random Resizing, Horizontal Flipping, and Photo Metric Distortion and a Poly learning rate schedule to ensure model robustness under varying lighting and road conditions.
- Achieved a peak Mean Intersection over Union (mIoU) of 0.6716 with the fine-tuned DeepLabV3+ model, outperforming the baseline by significant margins.
- Improved SegFormer performance by 27x over the zero-shot baseline, reaching an mIoU of 0.5341 through targeted fine-tuning.

Bird Species Detection

 [github/birdspeciesrecognition](#)

- Developed a fine-grained visual classification system using the Vision Transformer (ViT-B/16) architecture to automate the identification of 200 distinct bird species.
- Engineered a robust data pipeline by replacing default dataset splits with a custom stratified 80:10:10 split, ensuring proportional representation of minority classes and more reliable validation.
- Conducted an in-depth misclassification clustering analysis, revealing that 70.2% of errors occurred within the same bird family (e.g., Warblers, Sparrows), providing a roadmap for future hierarchical classification.
- Achieved a final test accuracy of 87.12% across 200 species on a verified, noise-free evaluation set.
- Reduced potential model bias by systematically surfacing and removing 4% of dataset label noise.

Inventurax: Sales & Billing Software

 [github/inventuraxpro](#) |  [inventuraxpro.vercel.app](#)

- Architected a lightweight, full-stack billing application designed for small businesses to perform instant invoicing and inventory tracking with zero setup cost.
- Developed a client-side PDF generation system using jsPDF and html2canvas, ensuring 100% layout consistency across all browsers by utilizing a canvas-capture approach.
- Integrated dynamic QR-based UPI payments by engineering a URI composer compliant with NPCI specifications, allowing for instant, pre-filled payment generation.
- Implemented secure session-based access and data persistence using Firebase Authentication and Firestore, enabling real-time multi-user synchronization.
- Developed instant invoice generation mechanism by eliminating server-side round-trips for document processing.

De Baker's & More | Full-Stack E-commerce Platform

 [github/debakersandmore](#) |  [debakersandmore.vercel.app](#)

- Architected a responsive full-stack e-commerce application for a bakery using **React.js** and **Bootstrap**, featuring a dynamic product catalog with real-time filtering and category-based navigation.
- Implemented a secure user authentication flow and integrated a **RESTful API** to manage dynamic data retrieval for products, pricing, and inventory updates.
- Optimized frontend performance by implementing **Lazy Loading** and image optimization techniques, reducing initial page load time and improving the overall user experience on mobile devices.

EDUCATIONAL QUALIFICATIONS

Gandhinagar Institute of Technology

Bachelor's of Technology in Computer Science

Affiliated by Gujarat Technological University

Oct 2021 - Mar 2025

- **Specialization:** Artificial Intelligence, Data Science, Database Management System, Operating System, OOP, Data Structures and Algorithms.
- **Major Projects:** Automated Smart Lighting System for Schools and Colleges, Medical Data Management System, etc.
- **Academic Excellence:** Secured a consistent CGPA of **9.03/10**, demonstrating strong foundational knowledge in Computer Science and Engineering.

Ahmedabad University

Master of Technology in Computer Science and Engineering

School of Engineering and Applied Science

July 2025 – Present

- **Specialization:** Artificial Intelligence, Data Science, Deep Learning, and Machine Learning.
- **Major Projects:** Hypoxic Ischemic Encephalopathy Lesion Segmentation, Road Marking Segmentation, Bird Species Detection, Hard Stop Detection, Diabetes Risk Prediction, etc.
- **Academic Excellence:** Currently maintaining a **3.26/4.00** CGPA through the first semester of the M.Tech program.

SKILLS

Languages

HTML, Python, CSS, JavaScript

Database & DevOps

Firebase, Git, MongoDB

Frameworks & Libraries

React, Bootstrap, Node.js, Transformers

Design & Creative

Figma, Photoshop, Premiere Pro